Generative Ai an Overview

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1. INTRODUCTION

The paper presents how does neural networks has changed to produce new data and the meantime to contribute as a compression technique. As the mnist present the reconstruction of images that display numbers as an initial approach. That each image have used as as a ground truth and generated data points have used for reconstruction. A dataset the contains ten thousand images, one thousand images for each digit.

As more modern approach a pretrained network have used to along with text analysis to reconstruct patterns that are prompt. The CompVis a project of Patrick Esser that makes use of an annotated dataset and it reconstruct an image from the ground according to what a user has already query. The network makes uses of pretrained weights (vqgan)imagenet. The implementation of a diffusion stable is the last approach that introduces a better compression than the autoencoders. Moreover as scientific approach on the loss functions that improves the network results and solves problems such as vanishing gradients. Diffusion models are new kind of neural networks that allows us to generat high quality images and sound, has show big interest to AI due to quality. The autoencoders and variational autoencoders, briefly autoencoder is a neural with high dimension at the input and the output but the centre is low dimension. The network has an encoder and decoder. The reconstructinos are not perfect.

The rest of the paper is organized as follows. In Section 4 how algorithms work and how we will use them for our experiment and which are the factors that we need to consider. In Section is our datasets, In Section 6 the evaluation of our experiment and finally in Section 8 conclusion and future work. Cited papers are referenced in the Section 9.

2. PROBLEM STATEMENT

It is not new that compression is of the most issues. As the number of data increase, the autonecoders has been widely used for such a purpose in the previous decade. In other words how it possible to shrink the size of a file and not to lose any information. The latent vectors of the autoencoders have been used store the more important information and when is need to reconstruct it . The approach takes as an input a number of data notated as (x). The data are feedforwarded to a network that introduces enough noise to on the file as they saved in the latent vectors, next the a decoder propagate to reconstruct the information. But is it possible to increase the compression? Another problem is how it is possible to that the reconstruction was correct and the new file is reconstructed correctly? Until recent approach the interpolation was a measure for images. A simple test that compare the reconstructed image with the one that was used to train the network.

3. AUTOENCODERS

Autoencoders was the fist approach to compress a file. However a more advanced approach of variational autoencoders use many concept from the field of bayesian machine learning and they inherently probabilistic in particular they introduce the constraint that shrink the vector to fit to a standard normal distribution. It's easy to generate new images we can only use the second half of the network the decoder. The whole process is base on normal distribution z N(0, 1), pass those to the decoder and draw an image hat(x) = decode(z). This unlike regular autoencoder, with a variational autoencoder we can do more about where the latent vector is, because we purposely train the network so that these vector follow a normal distribution.

4. GENERATIVE

Gans work very differently from autoencoder, the architecture consists of two neural networks, the generator and the discriminator that are training simultaneourly. The generator has as main process to create images the input of the generator is just a normal distribution z N(0,1) and produces the corresponding image G(z) = hat(x). The discriminator job is to differentiate real images and fake images created by the generator by training both network simultaneously we continue make both of them better eventually the discriminator becomes better to classify image from generator and the training dataset while the generator improves it's accuracy and can classify it's image as real. If generator is close to the state of art 50% accuracy.

To train of generator, begins the introduction of noise to the image notated Z, a mathematical vector made of random numbers, noise. It gives this as input to the generator architecture, that will produce an output that we going to call it hat(X). The hat(X), we give it as input to the discriminator, the discriminator produces an output Y that's the first step.

The loss function loss value expresses how good is the performance of the generator is.
In more details in a set of images the generated image is notated as 0 and the real image notated as 1. So when the image passes through the discriminator, at the begin it gain some value [0.1,0.7,0.9,0.3] those values are going to begin the prediction Y. The prediction the output of the discriminator, that compared with the actual label of the image. Then the loss value to use in the chain rule of calculus and the back propagation algorithm to use that to update the parameters, the internal parameters of the generator network. And this step keep repeating and repeating the same thing over and over. And that lost value ideally, is going to keep decreasing. To sum up the training phase , keeps update the parameters if the generator, thanks to the loss value. That is then back propagated back to those parameters. Next the traiin of discriminator is continue as follow it takes as input the real dataset that is trained and tested on the fake images produced by the generator.

4.1 loss function

The main method that the networks is cross-entropy, the cross entropy is the number of bits required to repesent an average event from one distribution compated to another distribution. So we have P that represents the target, the distribution. So the generator or discriminator, this is the targetm the ideal distribution P and Q is the distribution that is obtained through the training process. And the cross entropy is going to give us the number of bits required to represent an average event. The cross entropy is number of extra bits needed to represent an event using the apporiximation., the Q, instead of P.

$$H(P,Q) = -SUM_{i=1}^{n} P(x_i) log(Q(x_i)) \tag{1}$$

That is the cross entropy, the product, of the probability considering the target distribution and the logarithm of the probability considering approximation distribution.The entropy is used to measure the information and can correlated with the loss as the entropy high information has low value and low probability gives higher values.
For the case of the discriminator the loss is binary cross entropy loss equation. So the discriminator makes use of Y the real labels and the hat(Y)

To evaluate the fake images,it's gonna be logarithm of one minus the output of the discriminator applied to the output of the generator.

$$-1/nn^{S}UM_{i=1}(logD(x_i))log(Q(x_i)) \tag{2}$$

final equation that is applied that is composed of two parts binary one that deals with the real labels and one that deals with the fake images to calculate the lost value of the discriminator. The discriminator wants this first part to output of a one, because we want the real images to be predicted as real, as one, so logarithm of one is zero plus logarithm of one minus.

The discriminator want to predict more and more towards zero.

Discriminator Loss

$$BCELoss = -1/nSUM_{i=1}^{n} - (y_i logD(x_i) + (1-y_i) log(Q(x_i)) \tag{3}$$

the way you guide the network about how to tweak its parameters through back propagation is alos important.

5. DIFFUSION STABLE

The diffusion model continues with the same way using a generator and a discriminator. However the forward diffusion process we begin with an image as highly organised in a serious of steps we continouly add random noise from a normal distribution to an image after some steps that the noise has been added that the resulting image is look like noise in other it is highly disorganized. So the generative model takes the process in a reverse order and performs a series of steps in order to remove noise for an image. This backward diffusion process a highly disorganized image to an organize image.
Stable diffusion is a Latent Model because makes use of the latent to store the image. The process differs from the usual ones as it uses downsampling to store in the latent vector and upsampling for the reconstruction. The diffusion process in the centre has better performance than the already used techniques.

5.1 localization

The localization is has been widely used for generatives and prefered than object detection as the first refers to multiple objects and different locations. Especially for the dataset annotation. It is an object detection taks from image classification. With object localization the image corrensponds to the training data and the labels is the continous values that are coordinates on the (x1,y1& x2,y2) it is a regression task. To train a model to achieve object detection it makes use of the convolution layers that design a pattern and the height and width of an image decrease through each convolution because of the pooling layers.

6. EXPERIMENT

For this reasearch three dataset were used the first is the mnist dataset that handwritten digits reconstructed from the original. A generative model used with latent vector of 64 dimensions and trained in 300 iterations with binary cross entropy. The second approach is a pretrained network of imagenet and use prompts and images to generate image, the learning rate is 0.5 and total number of iterations 1000. At last a pretrained stable diffusion model for 50 iterations. For all the datasets it was measured the loss and the for the diffusion model and the vqmodel of imagenet the prompts.

7. RESULTS

As the result depicts the generatives started with an image that is presented as noise and not content. The first approach of mnist dataset starts with entropy close to zero value and the loss of the discriminator high in contrast with the latest iteration that the information was really low in the generator so the noise that expressed in the initial blank image was more than the first iteration and as the discriminator depict the average loss of the dataset improved [2]. The required time for training was more than the diffusion stables approaches. On the other hand the diffussion stable approaches

was both tested using prompts and the result of the imagenet vqgan (made by K. Crowson) is more abstract compared with the approach of the generative network of mnist and the vqmodel. The loss of the vqgan is approximately higher than the other networks and the weight of each phase was equal for the pattern reconstruction. At last the vqmodel that tested using same prompt with the vqgan network and time to execute the script was aproximately equal and the result depicts that the pattern was formed better than all the other approaches.
Further research on generatives with diffusion by fine tune the last two layers of the pretrained network [1]. The fine tune process does not concetrate on the process of discriminator but the construction of of noise with parameters beta_start=0.00085, and beta_end=0.012 in 1000 iterations to construct the noise and the learning rate stepped using cosine. The recostruction of the image was trained in just 50 steps and the observation was that the loss fluctuate from the starting point of 0.1295 to reach it's highest peak on an approximation of loss to 0.19 at 44 iteration to finaly reach a minimum of 0.1376. Pretrained of a dataset of specific genre that downloaded through the internet and using mean square error as a loss.

8. CONCLUSION

The paper has examine three approaches the generative ai and two diffusion stable approaches. A the outcome depicts the diffusion stable are not compared with the last one to provide a better outcome. The generative in three hundred epochs achieves the better performance in terms of loss for the reconstruction of the initial image however the diffusion stable that gives better compression and high quality results in less time seems to provide a better outcome especially for the vqmodel. The second approach does provide results that can be compared and this outcome might be the number of iterations that does not provide the image that was prompt, in contrast with less iteration the result was better but not comparable with other approaches. The diffusion stable models provide better reconstruction as the loss suggests.

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Network	Loss Generative	Loss Discrimin
Mnist generative 1st epoch	0.7386760061979298	0.66456787149
Mnist generative last epoch	4.13822067499161	0.08336465466
ImageNet [child,forest]		0.1298828125
ImageNet [child,forest]		0.24462890625
Stable diffusion	0.1356	

Table 1: Generative networks.

9. REFERENCES

[1] R. Gal, Y. Alaluf, Y. Atzmon, O. Patashnik, A. H. Bermano, G. Chechik, and D. Cohen-Or. An image is worth one word: Personalizing text-to-image generation using textual inversion, 2022.

[2] T. Mondal, L. E. T. T. Trang, M. Coustaty, and J. Ogier. Performance evaluation of deep generative models for generating hand-written character images. *CoRR*, abs/2002.11424, 2020.

Figure 1: z vector,initial image

Figure 2: the pattern

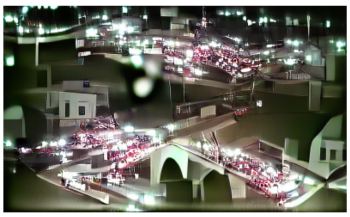

Figure 3: decoder image

Figure 4: vqgan imagenet 1000 iterations [child,forest]

Figure 5: vqmodel decoder

Figure 6: vqmodel decoder

Figure 7: vqmodel decoder

Figure 8: vqmodel 50 iterations [child,forest]